The numbers 0 and 1

01

In this sheet you will...

- see what happens when you add or multiply with 0 and 1
- use zero pairs: a number and its opposite
- understand why we cannot divide by 0
- use 0 and 1 to calculate faster

Key words

zero
one
opposite
zero pair

This sheet follows our lesson step by step, so you can read it again at home.

1 The number 0

Adding 0 changes nothing. Multiplying by 0 always gives 0.

Adding 0

$7 + 0 = 7$
$(-4) + 0 = -4$
$0 + 9 = 9$

Multiplying by 0

$7 \times 0 = 0$
$(-4) \times 0 = 0$
$0 \times 9 = 0$

A number and its opposite make a **zero pair**: $5 + (-5) = 0$ and $(-8) + 8 = 0$.

2 The number 1

Multiplying or dividing by 1 changes nothing. A number divided by itself is 1.

Multiplying by 1

$7 \times 1 = 7$
$(-4) \times 1 = -4$
$1 \times 9 = 9$

Dividing

$(-4) \div 1 = -4$
$7 \div 7 = 1$
$(-4) \div (-4) = 1$

Key idea **Adding 0 and multiplying by 1 leave a number unchanged.**

3 Why we cannot divide by 0

$12 \div 3 = 4$ because $4 \times 3 = 12$. Division asks: which number times 3 gives 12?

Now try $7 \div 0$. We would need a number so that $? \times 0 = 7$. But anything times 0 is 0, so no number works. That is why we say: **we cannot divide by 0**.

Watch out!

$0 \div 7 = 0$ is fine, because $0 \times 7 = 0$. But **$7 \div 0$** is not possible. Do not mix them up.
$7 \times 0 = 0$, not 7.

Exercise 39

1 True or false? If it is false, write the correct answer.

a) $8 \times 0 = 8$ .....

b) $-6 + 0 = -6$ .....

c) $0 \div 5 = 0$ .....

d) $9 \times 1 = 1$ .....

e) $4 \div 0 = 0$ .....

f) $7 + (-7) = 0$ .....

2 Calculate.

a) $(-12) \times 0 =$ .....

b) $(-15) \times 1 =$ .....

c) $0 \div (-4) =$ .....

d) $9 + (-9) =$ .....

e) $(-6) \div (-6) =$ .....

f) $23 \times 1 \times 0 =$ .....

3 Find the missing number.

a) $5 \times \square = 0$ $\square =$ .....

b) $\square + (-3) = 0$ $\square =$ .....

c) $\square \times 1 = -8$ $\square =$ .....

d) $0 \div \square = 0$ Which number can $\square$ not be? .....

4 Without a calculator, using 0 and 1 to help you.

a) $358 \times 0 + 12 =$ .....

b) $49 \times 1 - 49 =$ .....

c) $0 \times 15 - 3 =$ .....

5 Maya says: "6 ÷ 0 = 0, because there is nothing to share." What would you tell her?

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CHALLENGE

A calculator shows an error when you type 5 ÷ 0. Use the idea "? × 0 = 5" to explain why.

Summary checklist

- ☐ I know that adding 0 and multiplying by 1 change nothing.
- ☐ I know that multiplying by 0 gives 0 and that a zero pair adds to 0.
- ☐ I can explain why we cannot divide by 0.